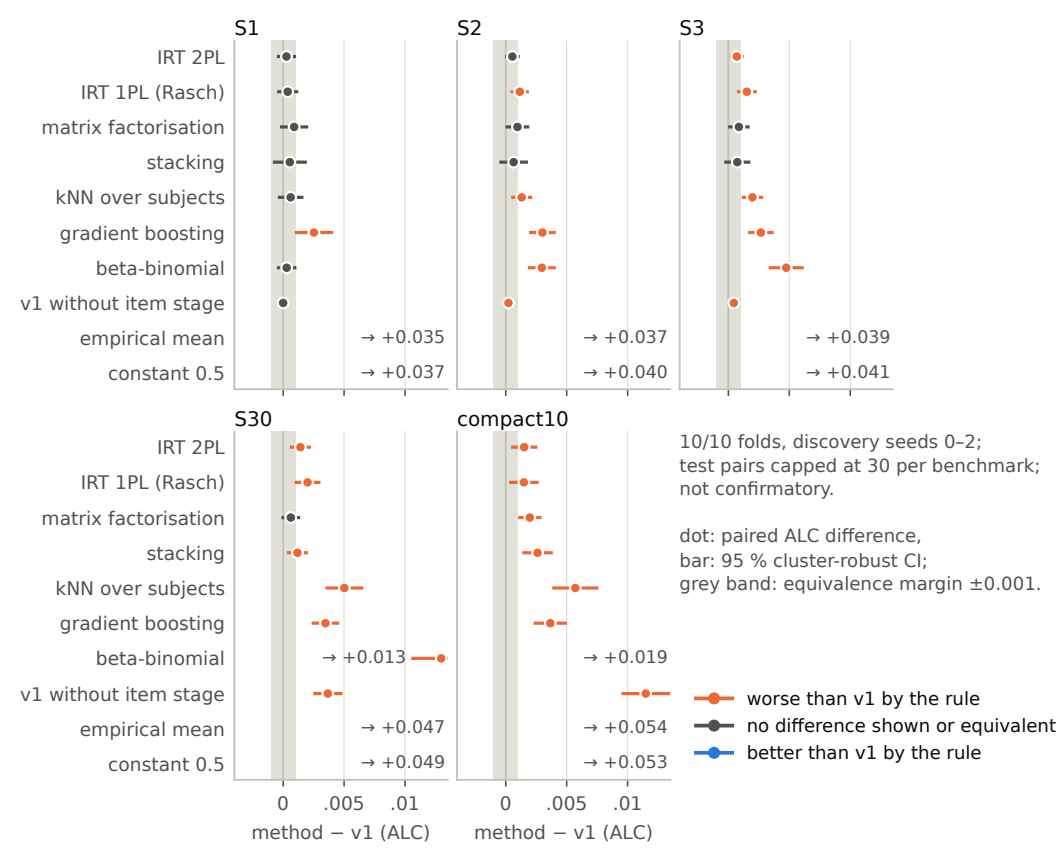
S1
S2
S3
S30
compact10
IRT 2PL
IRT 1PL (Rasch)
matrix factorisation
stacking
kNN over subjects
gradient boosting
beta-binomial
v1 without item stage
empirical mean
constant 0.5
→ +0.035
→ +0.037
→ +0.037
→ +0.040
→ +0.039
→ +0.041
→ +0.013
→ +0.019
→ +0.047
→ +0.054
→ +0.049
→ +0.053
0
.005
.01
method − v1 (ALC)
10/10 folds, discovery seeds 0–2;
test pairs capped at 30 per benchmark;
not confirmatory.
dot: paired ALC difference,
bar: 95 % cluster-robust CI;
grey band: equivalence margin ±0.001.
worse than v1 by the rule
no difference shown or equivalent
better than v1 by the rule